

9-2f Partial-Year Depreciation

A fixed asset may be purchased and placed in service other than the first month of an accounting period. In such cases, depreciation is prorated based on the month the asset is placed in service. For example, assume an asset is placed in service on March 1. For an accounting period ending December 31, depreciation would be computed (prorated) for 10 months (March 1 to December 31).

Assets may also be placed in service other than the first day of a month. In such cases, assets placed in service during the first half of a month are normally treated as having been purchased on the first day of *that* month. Likewise, asset purchases during the second half of a month are treated as having been purchased on the first day of the *next* month.

Straight-Line Method

Under the straight-line method, depreciation is prorated based on the number of months the asset is in service. To illustrate, assume that **Exeter Company** purchased the forklift on October 1 instead of January 1. The first-year depreciation would be based upon three months (October, November, December). First-year depreciation would be \$1,100, computed as follows:

$$\text{Annual Depreciation} = (\$24,000 - \$2,000) \div 5 \text{ years} = \$4,400$$

$$\text{First-Year Depreciation} = \$4,400 \times (3 \div 12) = \$1,100$$

Units-of-Activity Method

The units-of-activity method computes depreciation expense using an activity rate and the activity level for the period. To illustrate, assume that **Exeter** purchased the forklift on October 1 instead of January 1. Assume that during October 1 to December 31, the forklift was used for 400 hours. First-year depreciation would be \$880, computed as follows:

$$\text{Depreciation per Hour} = (\$24,000 - \$2,000) \div 10,000 \text{ Hours} = \$2.20 \text{ per Hour}$$

$$\text{First-Year Depreciation} = \$2.20 \text{ per Hour} \times 400 \text{ Hours} = \$880$$

Double-Declining-Balance Method

Like straight-line depreciation, if an asset is used for only part of a year, the annual double-declining-balance depreciation is prorated based on the number of months the asset is in service. To illustrate, assume that **Exeter's** forklift was purchased and placed into service on October 1 instead of January 1. First-year depreciation would be based upon three months (October, November, December). First-year depreciation would be \$2,400, computed as follows:

$$\text{Double-Declining-Balance Rate} = (100 \div 5) \times 2 = 40\%$$

$$\text{First-Year Annual Depreciation} = \$24,000 \times 40\% = \$9,600$$

$$\text{First-Year Partial Depreciation} = \$9,600 \times (3 \div 12) = \$2,400$$

The second-year depreciation would be computed by multiplying the book value on January 1 of the second year by the double-declining-balance rate. To illustrate, assume that Exeter purchased the forklift on October 1 and that \$2,400 partial depreciation was recorded on December 31. The book value on January 1 of the second year is \$21,600 (\$24,000 – \$2,400). The second-year depreciation would then be \$8,640, computed as follows:

$$\text{Second-Year Annual Depreciation} = \$21,600 \times 40\% = \$8,640$$

Business Insight

Tax Return Depreciation

The Internal Revenue Code uses the Modified Cost Recovery System (MACRS) to compute depreciation for tax purposes. Depreciation under MACRS is similar to that computed under the double-declining-balance method. MACRS is beneficial for tax purposes since depreciation is accelerated, thus creating faster tax deductions. Therefore, most companies use straight-line depreciation for their financial statements and MACRS for their tax accounting. This is a major reason for accounting and tax income to differ.